# AADI PENDSE

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## OBJECTIVE

Computer Science student at WPI with a strong foundation in Applied Mathematics, seeking Summer 2027 internships in Computer Engineering, Software Engineering, and Artificial Intelligence. Builds software ranging from low-level systems to full applications across Python, C, and data-driven architectures, with an emphasis on correct, well-structured code from numerical algorithms to robust, production-ready tools.

## EDUCATION

**Worcester Polytechnic Institute** — **Worcester, MA**

*Bachelor of Science, Computer Science (foundation in Applied Mathematics) | GPA: 3.9 (as of October 2026)* — *Aug 2025 – May 2029*

Activities: VEXU Robotics

**Hopkinton High School** — **Hopkinton, MA**

*GPA: 3.8* — *Graduated June 2025*

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, MATLAB, Swift, SQL, Racket, JavaScript

**Web & DevOps:** HTML, CSS, Git/GitHub, CI/CD pipelines, RESTful concepts, Serverless Deployment

**Core Concepts:** Data Structures & Algorithms, Systems Programming, Software Testing/QA, Object-Oriented Design

## COLLABORATION & LEADERSHIP

**VEX Robotics Club, WPI** — **2025 – Present**

*Build Team Member & Documentation Custodian*

- Support robot build activities while maintaining organized engineering notes on design decisions, build changes, testing results, and project progress.
- Preserve shared project knowledge to strengthen troubleshooting, design iteration, and cross-functional collaboration across mechanical, electrical, and programming teams.

**Independent E-Commerce Seller** — **Jun 2022 – Present**

*Self-Employed*

- Managed an end-to-end online business covering pricing analytics, product strategy, customer workflows, and inventory tracking.
- Maintained consistent quarterly sales growth while applying data-driven pricing models and digital marketing techniques.

## SELECTED PROJECTS

**Distributed Life-Safety & Special-Hazard Test Infrastructure** | PC Software, System Logic — **June 2026 – Present**

- Engineered inter-panel communication between a Notifier addressable master panel and a Potter conventional releasing panel, synchronizing multi-zone emergency response protocols across a distributed hardware architecture.
- Mapped cross-device state machines, asynchronous interrupt handling, and Boolean control logic to model real-time event propagation and fault isolation in distributed systems.
- Validated system boundary conditions and fault-tolerant behavior under electrical load models, applying software reliability principles to embedded hardware protocols.

**Serverless Web Platform & CI/CD Pipeline** | HTML/CSS, Git, Netlify — **July 2026 – Present**

- Designed and deployed a personal platform using a serverless JAMstack architecture for high-availability hosting.
- Configured automated CI/CD workflows via GitHub webhooks to trigger continuous integration builds on code pushes.
- Optimized web performance and security by configuring global CDN edge caching and SSL/TLS certificate management.

**Linear Algebra Algorithm Engine (RREF Solver)** | C — **May 2026 – Jul 2026**

- Implemented a C linear algebra solver executing Gaussian elimination to compute Reduced Row Echelon Form (RREF) for matrices.
- Optimized numerical stability and state transformations using memory management, bounds checking, and custom error handling.

**Academic Performance Tracking Tool** | Python — **Feb 2025 – Apr 2025**

- Developed a Python application to process academic data, compute weighted GPAs, and project performance trajectories.
- Engineered input parsing, structured exception handling, and edge-case validation to ensure execution reliability.

*Exceptional references available upon request.*